

Polynomial-size rational monotone models for the local comparison tensor

Research note; complete proof candidate

30 September 2026

Theorem 1. *There is an absolute constant C such that for every $m \geq 1$ there are an integer $K \leq Cm^2$ and rational numbers $p_{j,q}$, $1 \leq j \leq m$, $1 \leq q \leq K$, satisfying*

$$0 < p_{j,1} < p_{j,2} < \cdots < p_{j,K} < 1 \quad (j \in [m])$$

and

$$\frac{1}{K} \sum_{q=1}^K \prod_{j \in S} p_{j,q} = \frac{1}{|S| + 1} \quad (S \subseteq [m]). \quad (1)$$

Consequently the equal-weight product mixture with parameters $p_{j,q}$ has precisely the comparison tensor induced by a uniform random ranking:

$$\frac{1}{K} \sum_{q=1}^K \prod_{j \in S} p_{j,q} \prod_{j \notin S} (1 - p_{j,q}) = \int_0^1 t^{|S|} (1 - t)^{m - |S|} dt.$$

Scope. Thus rationality, positivity, monotonicity in the distinguished face, equal face weights, and *all* local comparison probabilities do not force an exponential number of faces. The exponential rational scalar-quadrature obstruction cannot be extended to these local hypotheses alone. A proof of an exponential per-die lower bound must use additional conditions involving the order among the other dice. This theorem does not produce a permutation-fair family.

The only quantitative input is the classical real Chebyshev-type quadrature bound: degree d admits an equal-weight interval quadrature with $O(d^2)$ nodes. For example, this is a special case of Gilboa–Peled, *Chebyshev-type quadratures for doubling weights*, Theorem 1.1, arXiv:1507.01505. The following lemma makes the necessary distinctness and interiority explicit, without assuming these additional properties in that theorem.

Lemma 2. *For every $m \geq 1$, there is a degree- m real equal-weight quadrature with K nodes satisfying*

$$m^2 \leq K = O(m^2), \quad 0 < u_1 < \cdots < u_K < 1.$$

Proof. Take a degree- $(2m + 2)$ real equal-weight rule with $K_0 = O(m^2)$ nodes, allowing repetitions and endpoints. It has at least $m + 2$ distinct support points: otherwise the square of the polynomial vanishing on its support would have degree at most $2m + 2$, zero quadrature sum, and strictly positive integral. At least m of the support points therefore lie in $(0, 1)$.

Repeat the entire rule $\lceil m^2/K_0 \rceil$ times. Its new cardinality K is at least m^2 and less than $m^2 + K_0 = O(m^2)$ (with a harmless nonstrict inequality when m^2/K_0 is integral). Select m coordinates at distinct interior support points as pivot coordinates. The Jacobian of the first m moment sums with respect to these pivots is an invertible scaled Vandermonde matrix. The analytic implicit function theorem therefore allows arbitrary sufficiently small perturbations of the other, free, coordinates, while uniquely adjusting the pivot coordinates to keep the first m moments exact.

All pivot coordinates remain interior and mutually distinct. Move each free endpoint slightly inward and choose the free perturbations generically to avoid collisions. Here is a precise verification that this is possible. Free-free collisions are proper affine hyperplanes. A pivot-free collision that holds at the starting rule is a nonzero analytic equation: varying only the colliding free coordinate, the derivative of its matching pivot is -1 , by the invertible Jacobian and the equality of their moment-derivative columns. Their difference thus has derivative -2 . Collisions not present at the starting point remain absent in a sufficiently small neighborhood. A finite union of proper real analytic zero sets cannot cover the open set of allowed inward perturbations. Sorting the resulting nodes gives the lemma. $\square$

Proof of Theorem ??. Choose a rule $u = (u_q)_{q=1}^K$ from Lemma ??. Choose pairwise disjoint sets $Q_1, \dots, Q_m \subseteq [K]$ of size m ; unused indices are permitted. Approximate u arbitrarily closely by a strictly increasing rational vector $t = (t_q)_{q=1}^K \in (0, 1)^K$. For $1 \leq s \leq m$, put

$$\mu_s(t) = \frac{1}{K} \sum_{q=1}^K t_q^s, \quad a_s(t) = \frac{1/(s+1) - \mu_s(t)}{s}.$$

For each j , solve the m -by- m Vandermonde system

$$\frac{1}{K} \sum_{q \in Q_j} f_j(q) t_q^{s-1} = a_s(t) \quad (1 \leq s \leq m), \quad (2)$$

and set $f_j(q) = 0$ outside Q_j . The solution is rational. Moreover it depends continuously on t near u , and it tends to zero as $t \rightarrow u$, because $a_s(u) = 0$ and the limiting Vandermonde matrices are invertible. Set $p_{j,q} = t_q + f_j(q)$. For a sufficiently accurate rational approximation, all columns remain strictly increasing and lie strictly between zero and one.

At every index q , at most one correction $f_j(q)$ is nonzero. Hence for $S \neq \emptyset$, writing $s = |S|$,

$$\prod_{j \in S} (t_q + f_j(q)) = t_q^s + t_q^{s-1} \sum_{j \in S} f_j(q).$$

Averaging and using (??) gives

$$\frac{1}{K} \sum_q \prod_{j \in S} p_{j,q} = \mu_s(t) + s a_s(t) = \frac{1}{s+1}.$$

The empty-set identity is automatic. Expanding the complementary factors $1 - p_{j,q}$ proves the claimed full comparison law. $\square$

Corollary 3. *The parameters in Theorem ?? can be realized by actual finite uniform dice: there are $m+1$ dice, one with $K = O(m^2)$ faces, for which all comparisons with that distinguished die have exactly the same joint law as under a uniform random ranking.*

Proof. Give the distinguished die ordered faces $q_1 < \dots < q_K$. For die j , assign its rational masses

$$p_{j,1}, p_{j,2} - p_{j,1}, \dots, p_{j,K} - p_{j,K-1}, 1 - p_{j,K}$$

to the $K+1$ gaps separated by those faces. Clearing denominators realizes these masses by equally likely faces; choose distinct labels for all faces inside each gap. Then $\Pr(D_j < q_q) = p_{j,q}$, as required. This assertion concerns only comparisons with the distinguished die. $\square$